

1. Implement Hyper-V

What is Hyper-V?

- Hyper-V is Microsoft's native hypervisor for creating and managing virtual machines (VMs) on Windows Server and some Windows 10/11 versions.
- It enables hardware virtualization, allowing multiple operating systems to run on a single physical machine.

Key Features:

- Support for generation 1 and generation 2 VMs
- Virtual networking, including virtual switches
- Support for virtual hard disks (VHD, VHDX)
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Live migration and replication (in Failover Clusters)

- Integration services for guest OS

Hyper-V Components:

- **Hypervisor:** The layer that abstracts hardware to run multiple VMs
- **Virtual Machine Management Service (VMMS):** Manages VM life cycles
- **Virtual Switch:** Provides networking for VMs
- **Virtual Hard Disks:** Disk images attached to VMs

Hyper-V Installation:

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Add the Hyper-V role on Windows Server via Server Manager or PowerShell (Install-WindowsFeature -Name Hyper-V -IncludeManagementTools).

VM Creation and Management:

- Create VMs via Hyper-V Manager or PowerShell
- Configure CPU, memory, storage, network adapters for VMs
- Use checkpoints (snapshots) to save VM states

2. Configure Disks and Volumes

Disk Types:

- **Basic Disks:** Use traditional partition tables (MBR or GPT)

- **Dynamic Disks:** Support volumes spanning multiple disks, RAID-like features (mirroring, striping)

Volume Types:

- Simple Volume
- Spanned Volume (across multiple disks)
- Striped Volume (RAID 0)
- Mirrored Volume (RAID 1)
- RAID 5 Volume (striping with parity, dynamic disks only)

Disk Initialization:

- New disks need initialization (MBR or GPT)
- Partition and format disks to create usable volumes

Disk Management Tools:

- Disk Management MMC snap-in
 - PowerShell (e.g., Get-Disk, Initialize-Disk, New-Partition)
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3. Implement Server Storage

Storage Spaces:

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Allows pooling physical disks to create virtual disks

- Supports resiliency options: Simple (no resiliency), Mirror, Parity
- Provides flexibility and easier management of physical storage

Storage Pools:

- Group of physical disks that form a single pool for allocating virtual disks

Storage Tiers:

- Using different disk types (SSD + HDD) for performance optimization

File Systems:

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NTFS (New Technology File System) – default Windows file system

- ReFS (Resilient File System) – designed for high resilience and scalability, commonly used in storage scenarios
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4. Implement Data Deduplication

What is Data Deduplication?

- Data deduplication is a storage optimization technique that eliminates duplicate copies of data to save disk space.

How it Works:

- Breaks files into small chunks (chunks)
- Stores only unique chunks

- Uses metadata to reconstruct files with references to unique chunks

Benefits:

- Reduces storage footprint
- Improves backup and restore times
- Saves bandwidth in replication

Supported Scenarios:

- Works best on file shares with redundant data (e.g., virtualization hosts, user shares)
- Available on NTFS and ReFS volumes

Enabling Deduplication:

- Enabled on file servers through Server Manager or PowerShell (Enable-DedupVolume)
- Configuration includes scheduling, file age limits, and exclusions